

LOHJA PORLA, Finland

WMO: 027060

Lat: 60.244N

Lon: 24.049E

Elev: 37

StdP: 100.88

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.0	(c) -17.4	(d) -23.2	(e) 0.5	(f) -20.7	(g) -19.8	(h) 0.7	(i) -17.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	26.4	19.5	24.6	18.3	22.9	17.3	20.8	24.4	19.5	23.3	18.3	21.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.2	22.9	18.1	13.1	21.2	16.9	12.1	20.1	60.2	24.4	55.6	23.2	51.9	21.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -22.3	(f) 27.9	(g) 6.1	(h) 2.1	(i) -26.7	(j) 29.4	(k) -30.2	(l) 30.7	(m) -33.6	(n) 31.9	(o) -38.0	(p) 33.4		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-22.5	21.8	6.0	1.6	-26.9	23.0	-30.4	24.0	-33.7	24.9	-38.1	26.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-5.4	-4.1	-1.3	4.5	11.3	15.0	18.1	16.5	12.1	6.0	2.4	-1.4
	DBStd	8.92	5.91	5.94	4.26	3.35	4.11	3.16	2.97	2.54	2.90	3.32	3.91	4.91
	HDD10.0	2146	478	395	349	167	32	2	0	0	14	128	228	352
	HDD18.3	4484	737	628	607	414	220	108	42	69	187	383	478	611
	CDD10.0	760	0	0	0	3	73	153	251	200	77	4	0	0
	CDD18.3	57	0	0	0	0	2	9	34	11	0	0	0	0
	CDH23.3	294	0	0	0	0	21	51	181	41	0	0	0	0
	CDH26.7	34	0	0	0	0	2	4	24	4	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	640	51	36	32	33	35	58	61	75	58	70	71	59
	PrecMax	829	92	104	65	61	74	100	156	162	138	188	175	151
	PrecMin	481	8	1	1	6	2	15	3	6	14	14	13	8
	PrecStd	89	23	25	15	17	20	23	37	36	29	38	32	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.2	9.5	18.2	25.9	27.1	29.0	27.1	20.4	14.1	10.4	8.6
		MCWB	3.6	3.1	4.9	10.4	16.4	18.5	20.8	20.4	16.7	12.5	9.6	7.6
	2%	DB	3.5	3.7	6.8	15.5	23.2	24.7	27.1	24.1	18.6	12.6	9.3	6.7
		MCWB	2.9	2.6	3.6	9.0	15.2	17.3	20.6	18.9	15.6	11.3	8.7	5.7
	5%	DB	2.5	2.7	5.1	12.9	21.1	22.9	25.5	22.2	17.3	11.7	8.1	5.4
		MCWB	2.1	2.1	2.7	7.6	13.9	16.4	19.5	17.5	14.9	10.7	7.5	4.6
	10%	DB	1.5	2.0	3.6	10.4	18.7	20.9	23.6	20.7	16.2	10.7	7.3	4.1
		MCWB	1.0	1.4	2.0	6.0	12.8	15.1	18.4	16.7	14.3	9.8	6.7	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.7	5.6	11.2	17.8	20.3	22.4	22.0	17.4	13.0	9.9	8.0
		MCDB	4.2	4.3	8.2	16.9	23.7	24.9	26.6	25.1	19.2	13.5	10.1	8.4
	2%	WB	3.0	2.8	4.1	9.5	16.1	18.4	21.4	19.6	16.4	11.9	8.8	5.9
		MCDB	3.5	3.4	5.9	14.6	21.1	23.2	25.3	22.7	17.9	12.4	9.2	6.6
	5%	WB	2.2	2.2	3.1	7.9	14.8	17.0	20.4	18.5	15.5	11.0	7.6	4.6
		MCDB	2.6	2.7	4.5	11.7	19.7	21.5	24.1	21.2	16.8	11.5	8.0	5.2
	10%	WB	1.1	1.5	2.2	6.6	13.3	15.8	19.1	17.5	14.6	9.9	6.7	3.5
		MCDB	1.4	2.0	3.3	9.9	17.8	19.8	23.0	19.9	15.7	10.4	7.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	6.6	8.1	9.1	7.8	7.6	6.8	5.6	4.7	3.3	3.8
		MCDBR	3.2	4.1	7.2	12.2	12.2	10.9	9.6	8.7	6.9	5.2	3.9	4.3
		MCWBR	3.2	3.5	4.6	6.8	6.2	5.5	4.6	4.2	4.3	4.1	3.9	4.2
	5% WB	MCDDBR	3.3	3.9	6.3	11.0	11.2	9.7	8.7	7.6	6.2	4.5	3.8	4.0
		MCWBR	3.4	3.4	4.5	6.6	6.1	5.4	4.6	4.4	4.3	4.1	3.8	4.0
Clear-Sky Solar Irradiance	taub	0.273	0.308	0.332	0.364	0.360	0.356	0.381	0.373	0.345	0.332	0.302	0.257	
	taud	2.129	2.190	2.261	2.342	2.416	2.464	2.418	2.435	2.505	2.400	2.216	2.072	
	Ebn at Noon	477	674	790	834	867	875	841	818	784	656	440	336	
	Edh at Noon	43	72	92	103	103	100	102	93	74	59	40	29	
All-Sky Solar Radiation	RadAvg	0.24	0.81	2.31	3.87	5.29	5.52	5.27	4.11	2.51	1.09	0.30	0.15	
	RadStd	0.04	0.16	0.29	0.40	0.55	0.43	0.49	0.42	0.32	0.17	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A

Nomenclature: See separate page